

CLASS 10 · CBSE · NCERT MATHEMATICS · FORMULA BANK

Chapter 7: Coordinate Geometry

Every formula and result of the chapter on one sheet

1	Distance between two points	$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
2	Distance from the origin	$\sqrt{x^2 + y^2}$
3	Section formula (internal, $m_1 : m_2$)	$\left(\frac{m_1x_2 + m_2x_1}{m_1 + m_2}, \frac{m_1y_2 + m_2y_1}{m_1 + m_2} \right)$
4	Section formula (ratio $k : 1$)	$\left(\frac{kx_2 + x_1}{k + 1}, \frac{ky_2 + y_1}{k + 1} \right)$
5	Mid-point	$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$
6	Centroid of a triangle	$\left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$
7	Point on the x-axis	$(x, 0)$
8	Point on the y-axis	$(0, y)$
9	Collinear points	$AB + BC = AC$ for some order of the three points
10	Parallelogram	Diagonals have the same mid-point
11	Points of trisection	Divide the segment in the ratios 1 : 2 and 2 : 1